

Product Manual

Kamoer Fluid Tech (Shanghai) Co.,Ltd.

Version: A / 1

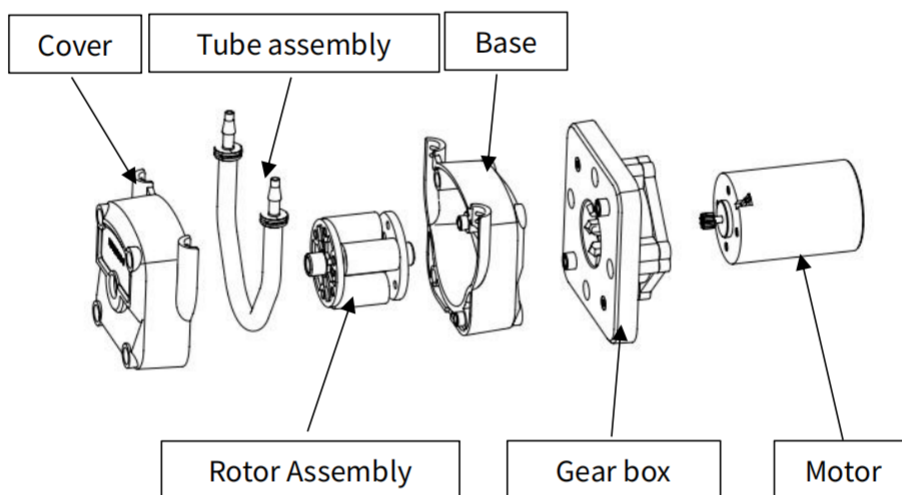
Product name	Micro peristaltic pump
Model	KPHM200
Release Date	2022.05. 23
Manufacturer	Kamoer Fluid Tech (Shanghai) Co.,Ltd.

A. Product Overview

1. Product Images



2. Product structure



3. Model definition

KPHM200	SV	null	B	16	null
Model	Motor model	Slow down	Tube material	Tube number	Installation method

3.1 Motor model



Brushed motor

SW 12V DC brushed

SV 24V DC brushed



Brushless motor

12B 12V DC brushless

24B 24V DC brushless



Stepper motor

ST 24V stepper motor



3.2 Deceleration

1: Level 1 deceleration, reduction ratio 1:8, 0: No deceleration (only stepper motor)

3.3 Tube type

B: BPT pump tube

3.4 Tube number

16#: 3.2mm×6.4mm

3.5 Installation method

Empty: standard installation

3.6 Product material

Pump head	Cover	Gear plate	Gear	Bracket	Connector
KPHM200	PA6	POM	PTE	PTE	PP

4. Performance characteristics, typical applications

4.1 Performance characteristics

◆ KPHM200 series peristaltic pump is a peristaltic pump driven by 3 motors, 24V/12V motor, which can provide different voltage options

◆ Pump tube selection - the pump tube is imported BPT (B) tube

◆ This peristaltic pump can be adapted to ordinary DC brush, DC brushless and stepper motors

◆ Compact structure, high cost performance

◆ Small installation method - the installation method is fixed through the plate and fixed by L-shaped sheet metal

◆ Transmission mode - the transmission mode is gear transmission, the primary reduction ratio is 1:8, high reliability and low noise.

4.2 Typical applications

Analytical instruments: ion chromatograph, five-class blood analyzer, hypochlorous acid water generator

Detergent Delivery: Automatic Car Wash Machines



Ion chromatograph



Auto car washer



Blood analyzer



Hypochlorous Acid Generator



5. Product Certification, Intellectual Property

Appearance design



Utility model



RoHS certification



CE certification

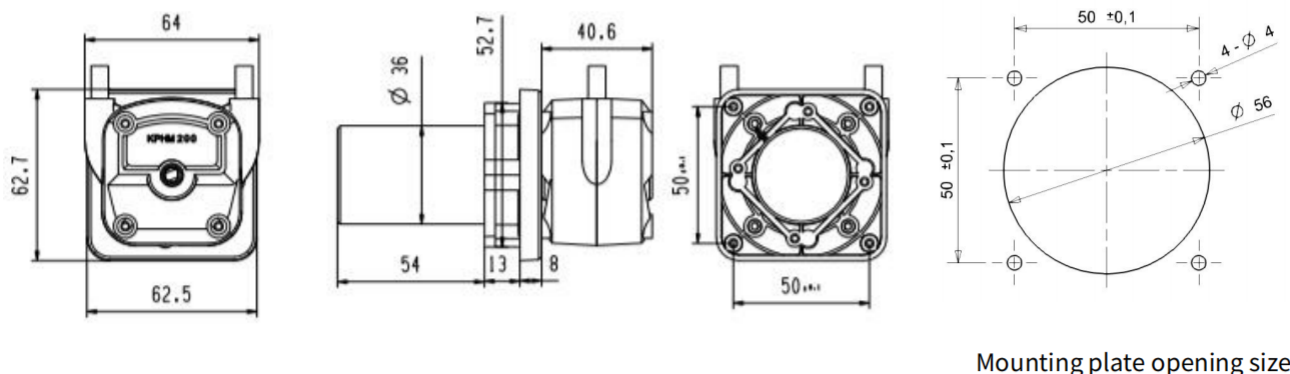


- ◆ Hoses are resistant to liquid media. Need to check chemical compatibility or conduct immersion tests; improper hose selection may cause hose damage quickly.
- ◆ The working environment of the product should not exceed 45°C, and the humidity should not exceed 70% (no condensation). The harsh working environment will cause premature damage to the product.
- ◆ Fluid leakage accidents due to hose ruptures depend on the fluid medium and your specific application conditions.
- ◆ High overloading may result in premature failure of the product.

B. Product specification

◆ KPHM200 DC Brush Product Specifications

1. Dimensions



2. Technical Parameters

Project		Technical Parameters
Basic parameters	Motor type	DC brushed motor
	Flow range	200-300ml/min
	Noise	≤65dB
	Power	10W
	Control method	switch control
	Weight	About 302g
	Rated voltage	12V/24V
	Pump tube life	BPT: 1000h
	Motor life	800h
Pump flow	Reference flow rate (unit: ml/min)	
	Pump tube code	B16 (1 6 #)
	ID × OD	Metric: 3.2mm×6.4mm Inch: 1/8 in×1/4 in
	Tube material	BPT
	SV/current: 0.3A SW/Current: 0.7A	≥200ml/min
Precautions	Remark	The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. The data are for reference only. In addition, it can be customized according to customer needs
	Reference noise value	In a quiet room with an ambient noise of 35 dB , the product is 50 cm away from the noise tester , and the noise is 58dB

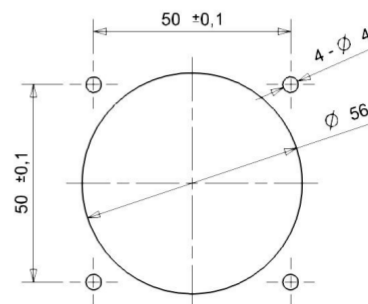
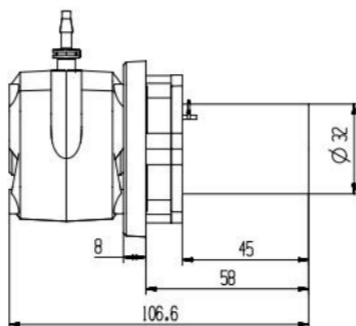
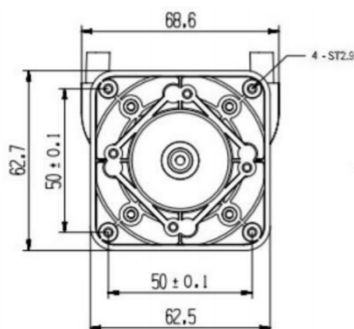
3. Motor control



Note: Connect the positive and negative electrodes according to the motor wiring diagram.

◆ KPHM200 Brushless DC Product Specifications

1. Dimensions



Mounting plate opening size



2. Technical Parameters

Project		Technical Parameters
Basic parameters	Motor type	DC brushless motor
	Flow range	200-300ml/min
	Noise	≤65dB
	Power	10W
	Control method	Switching value control/PWM speed regulation
	Weight	About 235g
	Rated voltage	12V/24V
	Pump tube life	BPT: 1000h
	Motor life	2000h
Pump flow	Reference flow rate (unit: ml/min)	
	Pump tube code	B16 (1 6 #)
	ID × OD	Metric: 3.2mm×6.4mm Inch: 1/8 in× 1/4 in
	Tube material	BPT
	24B/current: 0.3A 12B/current: 0.7A	≥200ml/min
Precautions	Remark	The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. The data are for reference only. In addition, it can be customized according to customer needs
	Reference noise value	In a quiet room with an ambient noise of 35 dB , the product is 50 cm away from the noise tester , and the noise is 57dB

3. Brushless Motor Wireless Correspondence Instructions

1. The red line is the positive pole of the power supply, the black line is the negative pole of the power supply, the white line is the speed control line, the yellow line is the signal feedback line (for speed measurement), and the green line is the commutation line (suspended to reverse, connect to the negative pole of the power supply for forward rotation).

2. Speed control port: PWM and 0-5V two speed control methods, speed range 13%-100% PWM frequency 10KHZ-30KHZ; amplitude: 5V, duty cycle 0-10% motor does not turn, 11%-100% Speed regulation range; 0-5V voltage speed regulation, 0-0.5V motor does not rotate, 0.6V-5V speed regulation range.

3. The speed feedback of the yellow line is connected to the oscilloscope probe or the host computer. 1 pulse/revolution.

4. Run at full speed, connect the red and white wires to the positive pole of the power supply, and the black wire to the negative pole

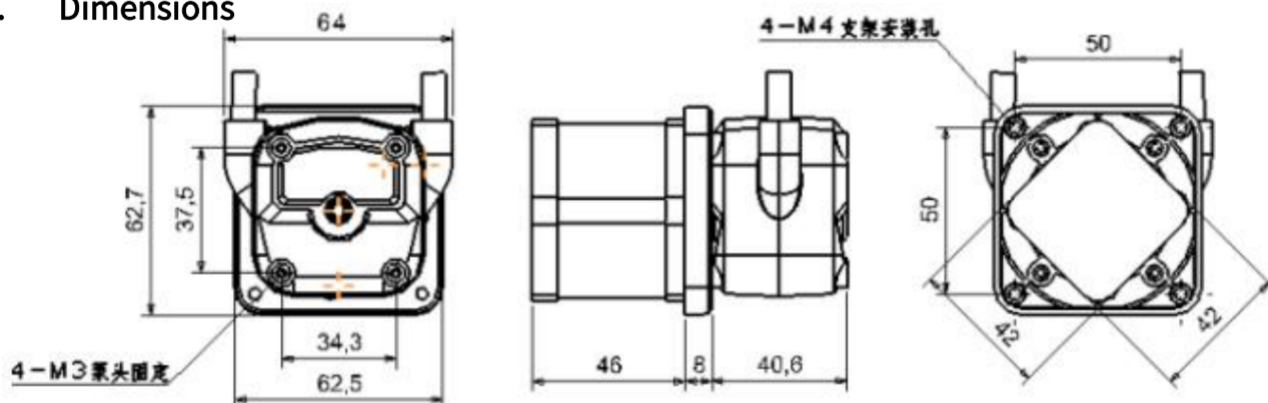
UL1332 Material: PTFE



- 1- FG-24AWG-yellow signal feedback
- 2- Vcc-24AWG-red positive power supply
- 3- SP-24AWG-white speed port
- 4- GND-24AWG-black Negative pole of power supply
- 5- F/R-24AWG-green reversing port

◆ KPHM200 Stepper Product Specifications

1. Dimensions

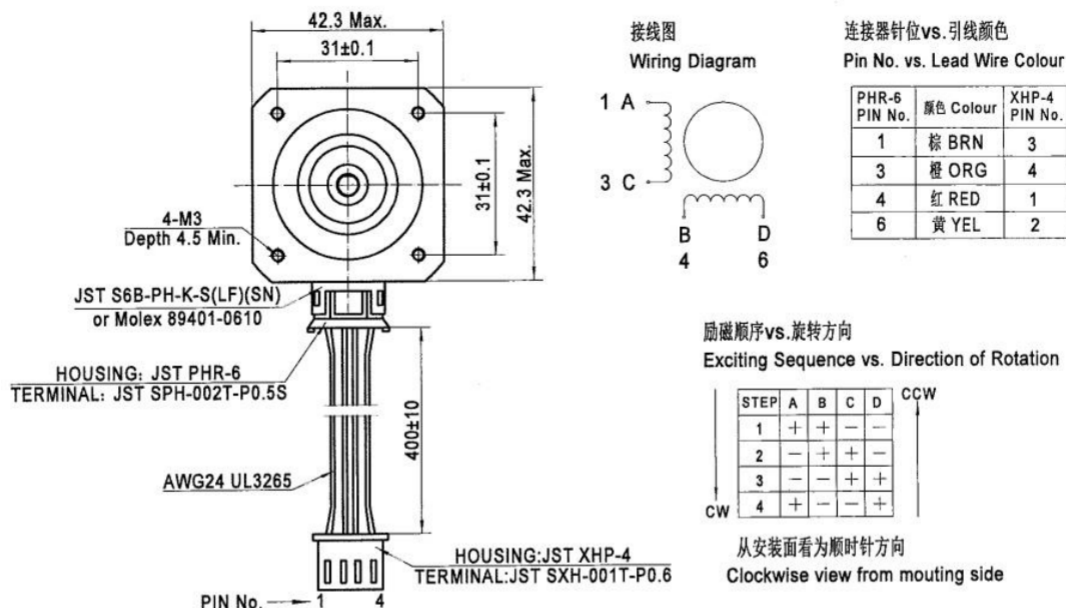


2. Technical Parameters

Project		Technical Parameters
Basic parameters	Motor type	Stepper motor
	Flow range	0-300ml/min
	Noise	≤ 60dB
	Power	20W
	Control method	Stepper driver
	Weight	About 458g
	Rated voltage	24V
	Pump tube life	BPT: 1000h
	Motor life	6000h
Pump flow	Reference flow rate (unit: ml/min)	
	Pump tube code	B16 (1.6 #)
	ID × OD	Metric: 3.2mm×6.4mm Inch: 1/8 in × 1/4 in
	Tube material	BPT
	400 rpm	220ml / min
Precautions	Remark	The above parameters are measured with pure water without pressure at room temperature and standard atmospheric pressure at 20°C. Actually, there will be certain errors in the flow rate due to different media, different outlet pressures, and DC motor speed errors. The data are for reference only. In addition, it can be customized according to customer needs
	Reference noise value	In a quiet room with an ambient noise of 35 dB , the product is 50 cm away from the noise tester , and the noise is 55dB



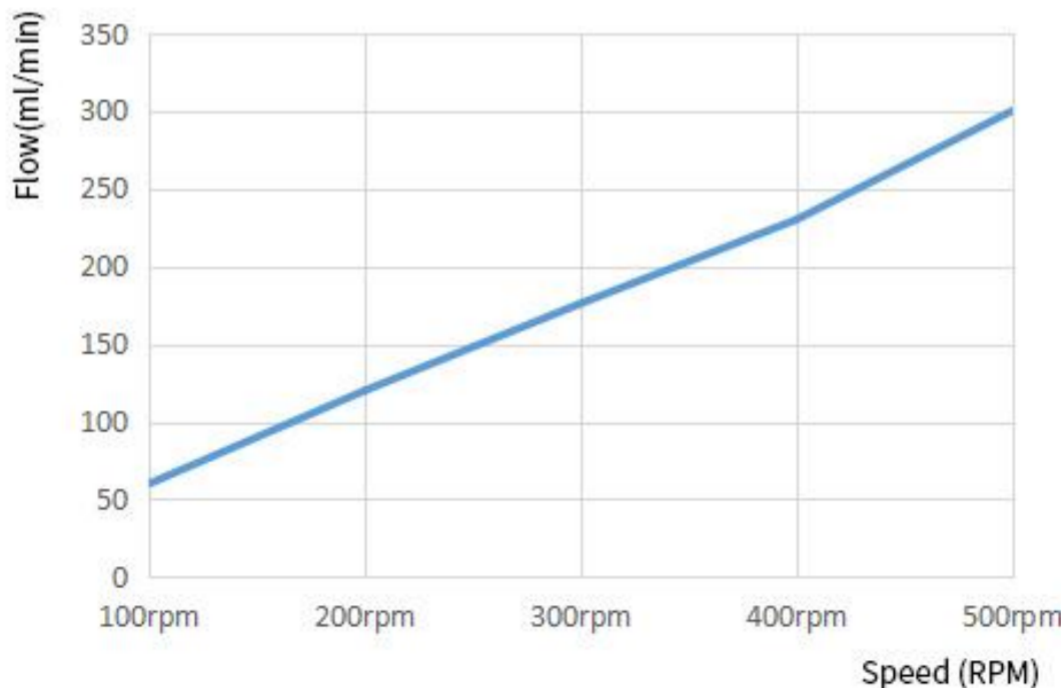
3. Motor parameters and line color diagram



Phase	Resistance	Inductance	Current	Holding torque	Insulation class
2	3.1±10%Ω	4.9±20% mH	1.2A	0.54Nm	B

4. Flow curve

KPHM200 Flow Curve



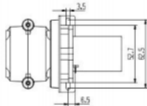
Flow curve is for reference only, and the specific parameters are subject to the actual working condition test.



C. Adapter tube

Pump tube material	Characteristic	Standards compliant	Tube number
BPT tube (B) 	Resistant to common chemicals, acid and alkali, and oxidation resistance, with a lifespan of more than 1000h	RoHS, FDA	16#

D. Optional accessories

Name	Specifications		Function introduction
L shape mounting plate	70.3 × 69.5 × 15mm		Add installation methods to meet different customer needs
Straight connector	1/8		3.2mm × 6.4mm hard tube or similar hose can be connected to the external
Peristaltic pump stepper drive	MODBUS-RTU		Can realize built-in and external potentiometer speed regulation, can realize MODBUS RS485 communication and 0-10V analog speed regulation, realize electric and trigger two start and stop modes, subdivision maximum 32, drive current can be set (0.7A, 1.2A, 1.7A, 2.2A, 2.7A, 2.9A, 3.2A, 4.0A)
Pulse generator board + stepper driver board	2802 + KMD-42B-P		Support clockwise and counterclockwise operation, LED 4-digit digital tube display, various control methods: rotary encoder, foot switch, external analog, 485 communication control

Special declaration: The company reserves the right to improve the appearance and technical specifications of the product without prior notice.

